

SAIQ, Oman

WMO: 412540

Lat: 23.076N

Lon: 57.643E

Elev: 1986

StdP: 79.63

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.6	5.0	-23.8	0.6	11.7	-19.7	0.8	12.4	10.1	10.8	8.9	11.4	1.9	20	0.356

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.2	31.8	13.7	30.9	13.7	30.1	13.7	18.2	25.0	17.5	24.9	16.8	25.0	3.7	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.8	14.4	22.8	14.7	13.4	22.8	13.6	12.4	22.0	60.4	24.8	57.7	24.6	55.1	25.0	21.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.0	7.6	6.5	1.1	33.2	1.8	0.9	-0.2	33.9	-1.3	34.4	-2.3	34.9	-3.7	35.6		
			-3.4	19.7	1.2	0.9	-4.3	20.3	-5.0	20.9	-5.7	21.4	-6.6	22.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.8	10.7	12.4	15.6	19.1	22.9	25.9	26.1	25.1	22.8	19.0	14.4	11.6
	DBStd	5.89	2.53	2.75	2.34	2.16	2.13	1.92	1.54	1.55	1.60	1.87	1.91	2.43
	HDD10.0	47	22	11	1	0	0	0	0	0	0	0	0	13
	HDD18.3	851	237	168	87	17	0	0	0	0	14	118	209	
	CDD10.0	3268	44	77	175	273	398	477	498	467	384	281	132	63
	CDD18.3	1031	0	1	3	40	141	227	239	209	134	37	0	0
	CDH23.3	7597	0	1	4	99	926	2158	2159	1504	656	90	0	0
	CDH26.7	2007	0	0	0	1	195	712	658	371	71	0	0	0
Wind	WSAvg	2.7	2.4	2.8	3.0	3.2	3.0	3.2	2.5	2.9	2.7	2.5	2.1	2.1
Precipitation	PrecAvg	199	12	20	29	23	13	10	28	22	15	10	9	9
	PrecMax	645	82	112	334	74	72	61	129	80	87	56	62	43
	PrecMin	76	0	0	0	0	0	0	4	0	0	0	0	0
	PrecStd	108	23	25	63	25	15	14	30	20	18	16	15	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.6	21.5	24.1	26.4	32.5	33.3	32.5	31.7	29.2	26.3	21.7	19.9
		MCWB	6.8	9.0	8.8	11.3	13.1	14.1	14.3	14.3	12.9	11.3	10.3	6.6
	2%	DB	18.1	19.9	22.7	25.4	29.9	32.1	31.5	30.4	28.3	25.4	20.6	18.6
		MCWB	7.0	8.2	8.8	10.9	12.4	13.5	14.3	14.4	12.9	11.2	10.1	6.8
	5%	DB	16.8	18.7	21.6	24.6	28.7	31.1	30.7	29.6	27.5	24.3	19.8	17.6
		MCWB	7.1	7.9	8.8	10.6	12.3	13.0	14.5	14.4	12.8	11.0	10.0	7.0
	10%	DB	15.5	17.5	20.5	23.7	27.4	30.2	29.8	28.8	26.7	23.3	18.9	16.4
		MCWB	6.6	7.8	8.7	10.3	12.0	12.8	14.4	14.4	12.5	10.7	9.5	7.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.7	12.2	13.6	17.0	16.5	17.5	19.2	19.0	16.7	15.5	14.4	11.6
		MCDB	14.1	17.9	17.6	19.6	23.5	25.7	26.3	25.2	23.3	19.9	18.6	14.8
	2%	WB	9.9	11.1	12.3	13.9	15.4	16.2	18.3	18.2	15.9	14.4	12.6	10.8
		MCDB	13.6	15.7	17.0	19.1	22.9	25.5	25.6	24.4	23.0	19.3	17.2	14.1
	5%	WB	9.2	10.2	11.4	12.9	14.7	15.4	17.6	17.5	15.2	13.5	11.7	9.9
		MCDB	13.2	15.0	16.9	19.6	23.1	26.1	25.4	24.4	22.9	19.3	16.7	13.6
	10%	WB	8.5	9.4	10.5	12.1	13.9	14.7	16.9	16.8	14.5	12.5	11.0	9.1
		MCDB	13.0	14.8	17.2	20.8	23.6	26.6	25.5	24.7	23.1	19.8	16.4	13.4
Mean Daily Temperature Range	5% DB	MDBR	7.7	7.9	7.9	7.6	8.1	7.8	7.2	7.1	7.3	7.5	7.6	7.6
		MCDBR	9.0	8.9	8.6	8.1	8.7	8.1	7.7	7.4	7.7	7.9	8.1	8.9
		MCWBR	6.0	5.5	4.3	3.6	3.5	3.6	3.8	4.0	4.2	3.8	4.3	5.6
	5% WB	MCDBR	7.2	7.3	6.9	7.0	7.8	7.4	7.1	6.7	7.2	6.8	7.1	6.5
		MCWBR	5.6	5.5	4.7	4.3	4.2	5.0	4.6	4.5	4.7	4.1	4.6	4.9
Clear-Sky Solar Irradiance	taub	0.306	0.335	0.377	0.422	0.446	0.503	0.569	0.522	0.435	0.354	0.324	0.305	
	taud	2.475	2.357	2.184	2.065	1.990	1.857	1.734	1.838	2.045	2.301	2.424	2.470	
	Ebn at Noon	965	953	922	883	854	800	749	787	854	919	935	951	
	Edh at Noon	99	119	148	170	182	207	234	210	168	124	103	96	
All-Sky Solar Radiation	RadAvg	4.69	5.48	6.34	7.00	7.58	7.41	6.91	6.77	6.42	5.86	4.89	4.43	
	RadStd	0.21	0.24	0.34	0.37	0.16	0.27	0.28	0.17	0.12	0.16	0.13	0.20	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (13 neighbors)	+0.42	+0.68	N/A	N/A	+0.28	+0.30	N/A	N/A	+96	+98	

Nomenclature: See separate page